CL Syllabus

Lydia's questions

What lens would you give in each situation and why

Situation	Options	Justification
Spherical Rx, Spherical Cornea	Soft Sphere	Simpliest fit
	Spherical RGP	Less inflammatory events, better optics.
Toric Corneas/Rx <2.00DC Corneal Cyl	**First line:** toric soft contact lens	Simple fit, chance it may stabilize.
	Sphericcal RGP <1.00DC = Flat K 1-2.00DC = Flat K + 1/3ΔK	
Toric Corneas/Rx >2.00 Corneal Cyl	Bitoric lens	High toric cornea needs a back surface lock. The lab will work out the measurements for you
Refractive ≠ Corneal Cyl	Bitoric CPE - some internal	Front correction corrects internal cyl so needs a strict
Refractive = corneal cyl	Bitoric SPE – no internal	Front correction only corrected the induced cyl, so lens can rotate to pleasing
Toric cornea/Rx >2.00 corneal Cyl CC(1.2-1.4) = RC	BST toric Using the BST induced to correct the cyl	With the refractive cyl being bigger then the corneal cyl by a factor of 1.2 to 1.4, it allows us to use the induced Cyl
Special case Spherical cornea Toric Rx = all internal cyl	Front surface toric	Having a spehical back means you can correct all internal with the front surface
	Soft toric lens	Similarly with a soft CL you can add the power and the astigmatism will be transferred to the back

Dynamic

Lens **centration**

Lens **movement**

Lens and **lid**

Fitting

Lens **cornea** fitting relationship

Descriptors: Loose and tight= dynamic Steep and flat = lens-corneal

Area 1: Myopia and functional contact lenses

Myopia Miserable, Hyperopes Happy

- When a myope wears their spectacles they are getting a base in prism when converging. Therefore when you take this prismatic effect away, a contact lens wearer will have to converge more.
- Through contact lenses, myopes accommodate more. This is because the vergence at the corneal plane experiences a slight difference leading to more accommodation required by the myope to be greater now that there's a lack of vertex distance between the divergence lens and the eye.
- Because of these increased binocular vision demands, an insipient presbyope may reach a "tipping point" in their reserves and experience prominent symptoms and blurry vision

Possible problems:

- Insipient presbyopia precipitated due to increased accommodative demand.
- Convergence insufficient patient who is borderline
- Accommodative insufficient patient
- *Patient was not vertex corrected.*

Solutions for these situations:

- Vision training e.g. brock string if convergence problem
- Supplementary reading glasses with or without prism
- Multifocal SCL with low add for the presbyopia
- Encouragement with younger/consistent wear patients to adapt.

Why would high myopes like CI's more than glasses

- Greater field of view No frame scotoma
- Less minification of vision
- Better cosmesis
- Less weight on frames
- Superior optics off centre – less aberrations and distortions in shape
- Ability to play sport
- Ring diplopia

Area 2: Keratoconus

Signs of keratoconus:

- Decreased VA
- High cylinder
- Oil droplet and scissor reflex
- Iron deposits and flesicher's ring.
- Hydrops (break in descemets causing hydro influx)

Associations:

- Ehler danlos
- Down syndromes
- Marfans
- Osteonesis imperfecta.
- Vernal keratoconjunctivitis

Types of keratoconus

- Early/Forme fruste, slight corneal distortion and no progression, manage with glasses. Onset would be from any age. Much more mild and is not as progressive. **Spectacles is ok.**
- Moderate = use RGPs or miniscleral lenses. Look at kerasoft lenses.
- Advanced = same as moderate but steeper, needs a vault.
- Severe: substantial scarring and thinning, poor vision even with RGPs and even lens intolerance. Needs grafts.

Onsets:

- Puberty onset – during teenage years, progresses for 8-10 years (mid 20s)
- Late onset – late 20s early thirties, less severe

Keratoconic general management

- Look for the signs of keratoconus
- Refer for cross linking
- Think about treatment options in regard to condition stage. (as above)

Staging and disease progression of keratoconus

- Condition is bilateral and assymetrical - 50% of normal fellow eyes will develop KC in 16 years.
- Highest axis of corneal power: Mild <48D, Moderate = 48-54, Severe >54
- Moderates can't get 6/6 behind a phoropter

Contact Lens Options:

- Rose K2: a lens that has the posterior shape that fits to the cone of the eye it has a smaller optic zone to allow for a even distribution of tears
- Kerasoft
- High modulus soft e.g. PureVision (1.1) or N+D (1.5) or even proclear

What is irregular astigmatism

- Where the principle meridians are not 90 degrees apart, the implication of this is that it cannont be corrected with simple correctable lenses. If the irregular astigmatism source is in the cornea, we can use the tear prism to fill in and create an even surface with the posterior lens surface.

Imaging and keratconus:

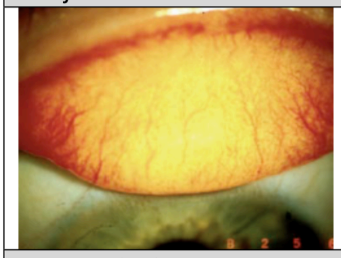
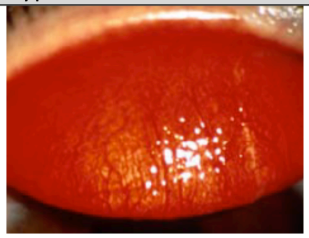
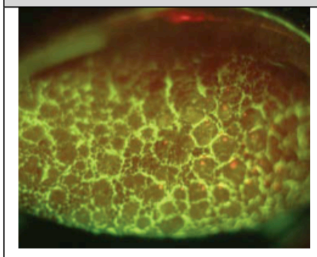
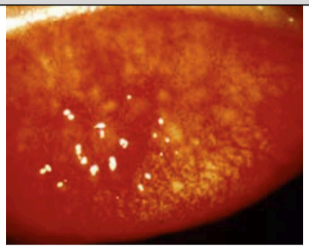
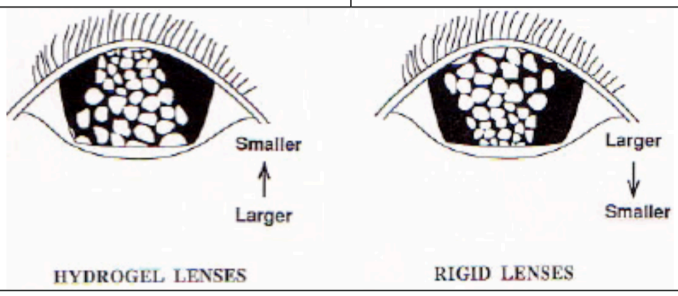
- Tomography is becoming the preferred method for keratoconus detection due to the posterior changes becoming more apparent in keratoconus before front surface changes. It does this by comparing it to a ellipsoid.
- Posterior changes more obvious then anterior changes
- **Do your E-300 before you put any drops in, tear film stability plays a big part.**
- Screening = use the axial dioptre corneal power display.
- Look for a:
    - A**ssymetric bow tie**
    - Pachymetric difference between 2 eyes. → 23um difference = 3 SD
- If they're wearing CL → take 2-4 weeks for the effect of CLs to dissipate

Topography v.s. tomography

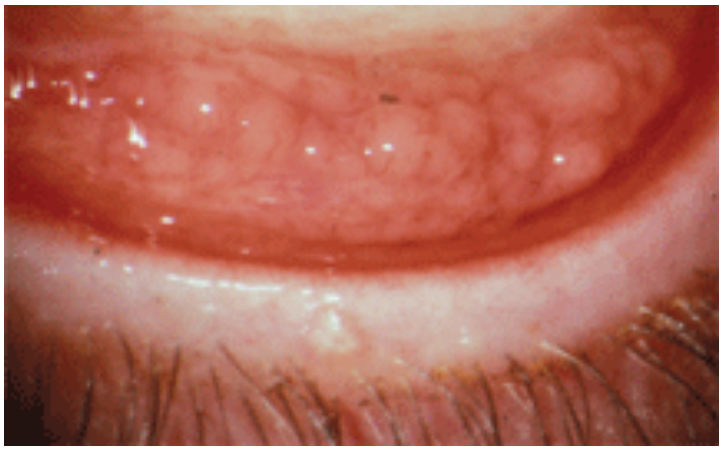
- Topography faster, easier to see corneal irregularities, the mires will show it clearly. Placido methods can be as much as 20 times more sensitive to corneal problems
- Topography is less affected by blink and is less affected by reflections
- Topography doesn't factor in the posterior surface
- Topography doesn't get affected by corneal opacities, still get a front map
- Topography can do eye steering
-

Keratoconus 3 strikes:

- Pachymetry map is thin, assymetrically thing. How thin is too thin on pachymetry maps. 3% significant, 6% absolutely.
- Posterior float map that compares the cornea to ellipsoid
- If in same location and 10um → patient has keratoconus, no LASIK.

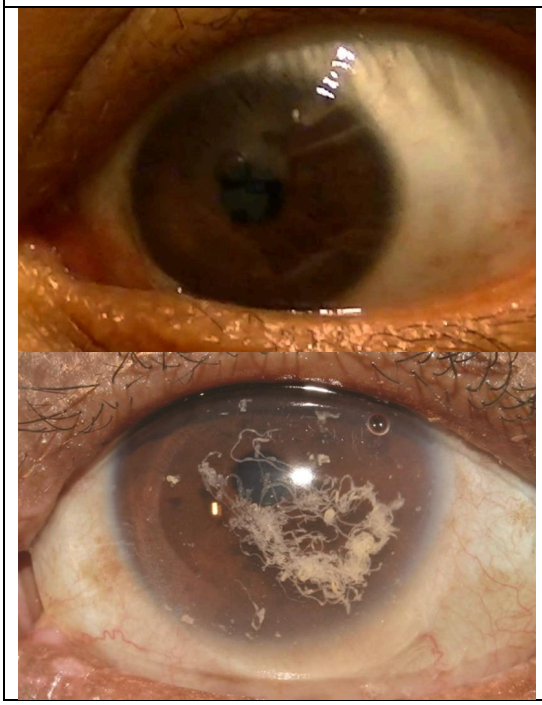
Normal Palpebral Conjunctiva	Papillary Conjunctivitis	Palpebral Conjunctival hyperaemia	Giant Papillary Conjunctivitis (GPC)
			
CLPC with Fluorescein	Local CLPC		
			

Area 3: Slit lamp findings

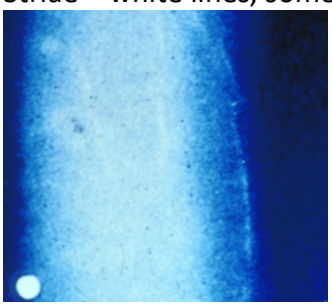
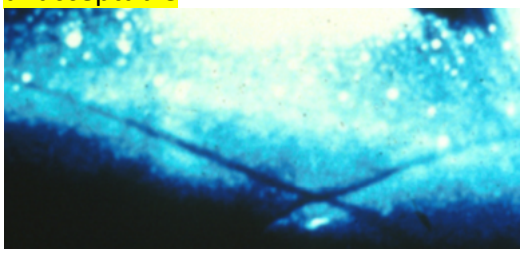


Left: papillae. Right: follicles
Papillae is a vascular change with surround edema. At the middle of each mound is the termination. Follicles is conjunctival lymphoid clusters of lymphoid tissue indicative of viral.

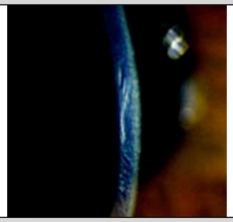
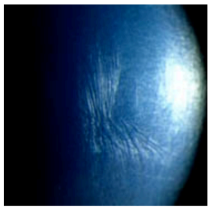
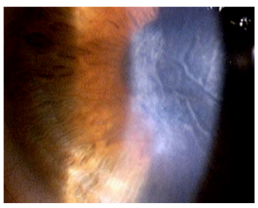
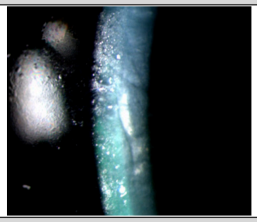
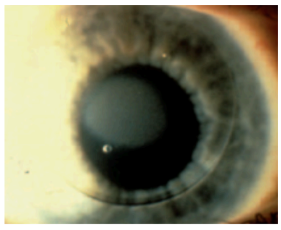
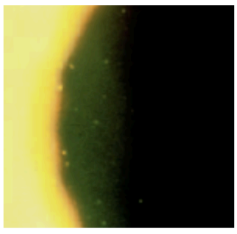
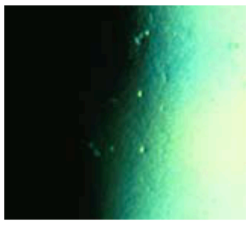
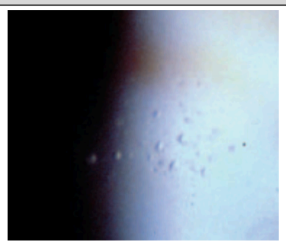
What does a mucus strand look like v.s. deposits

Mucus	Deposits
Mucus is lipophilic and more common within silicon hydrogel or RGPs	More common in ionic lens because they're more hydrophilic
Caused by accumulation	Caused by accumulation
Removing mucus strands can irritate the eye because strands anchor to the ocular surface.	Deposits that aren't removed can cause CLPC. Deposits can denature and cause reactions.
	

Striae and microcysts

Microcysts and vacuoles	Striae and folds
 Normal lighting, water low index in vacuoles Microcysts apoptotic material, higher index Related to hypoxia definitely Oxygen standards: 24 to avoid day time edema 34 to limit overnight edema to 8% daytime recovery 87 to limit to 4% → equivalent to no lens edema 125 to limit to 3% → updated of the above.	More related to edema due to buckling of the posterior stroma but edema as we know is related to hypoxia. Striae = white lines, some can be expected however  1 = 5% 10 = 11% (unacceptable) Folds = black lines, definitely >10% edema and clinically unacceptable  1 = 8%, 10 = 14% (all unacceptable)

Corneal Hypoxic Response

Striae	More Striae	Folds	More Folds
			
Central Corneal Clouding (epithelial oedema) - RGP	Epithelial Microcysts	More Epithelial Microcysts	Epithelial Vacuoles
			

Acute Epithelial Hypoxic changes: **intracellular** fluid accumulating, **lipid** like bodies in cells, **distortion** of cells, loosening of **tight** junctions and ⇓**transcorneal** potential, reduced **mitosis,** reduced **corneal sensitive**.

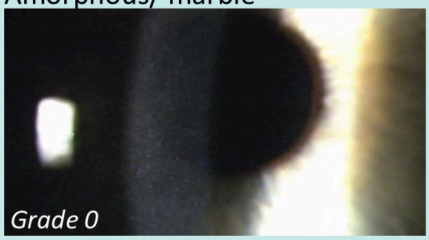
Chronic Epithelial changes: **thinning**, reduced oxygen **uptake**, **microcysts** and **vacuoles**, more cell retention→ more **fragile** cells, less **slach** syndrome → less **adhesion**,

Chronic stroma stromal: thinning, conceals edema. Haze, ECF, **increased stromal nerves,**
Acute stroma: stromal acidosis → CO2… increase the O2 and you get CO2 increase.

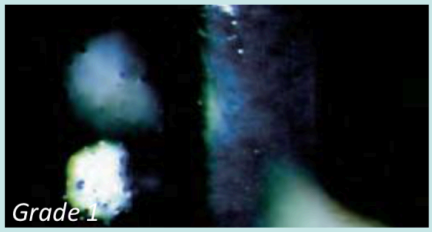
Endothelium, edema blebs, guttata, polymegethism → no evidence of anything yet.

Different tear film apperances

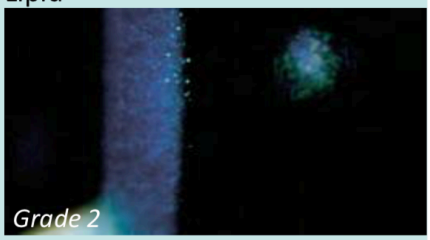
Amorphous/ marble



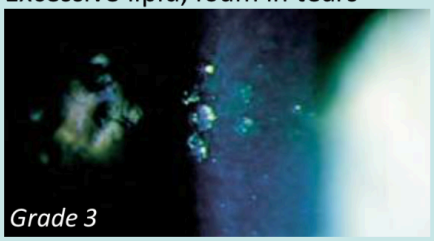
Polluted tear film



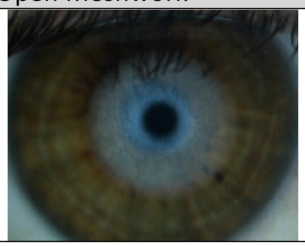
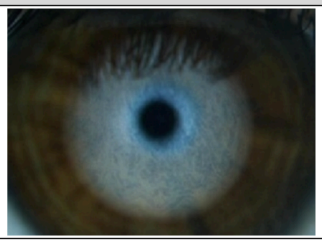
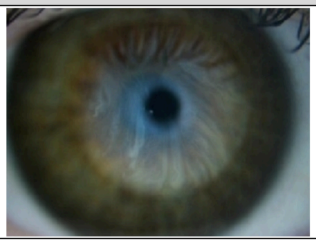
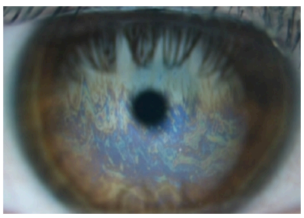
Lipid



Excessive lipid, foam in tears



No lipid layer

Tear Film Assessment			
Open Meshwork	Closed meshwork	Wave	Colour Fringes
			

Use a lower intensity to avoid bleaching and use a higher mag

Other Assessment gradings.

1a= open meshwork = not filled (marble like)
1b = closed meshwork = still got some gaps
2 = amorphous = harder to see, can see debris
3 = flow pattern = flow line for lipid
4 = colour fringes = too much oil, can destabilize, bad

Dry Eye Measurement:

- TBUT – said to be 10 seconds, I don't believe this.
- Tear prism = 0.25 = high, 0.1mm = low. But if 0.01 different between eyes = questionable.
- Cotton thread 15seconds → 15mm (should be anywhere between 9 and 20)
- Shirmer test 5 minutes → 10mm
- OSDI/DEQ/McMonnies.
- Keratometer
- Tearscope

Area 4: Lens dynamics and fitting

Steep/flat lens what happens to VA after a blink and why?

- Tight fit/steep lens = vision clears after blink → steepened lens has plus power, blink flattens lens to confirm better.
- Loose fit/flat lens = vision worsens after blink → moved the lens off the optic centre

Fitting dynamics → what if a lens is too tight/loose, how is it bad?

Too tight	Too loose
Improves on blink	Worsens on blink
Limbal trauma, indentation trauma, blood flow restriction. Lens adhesion	Lid trauma, limbal trauma, lost lens/decentration
Initial comfort but over time uncomfortable	Immediate discomfort.

Cornea power crosses

Corneal Cyl + Residual Cyl = Total or Rx Cyl

7.85mm @ 90, **7.9** @180 → corneal cyl = -0.25**x180**
(get the higher number, that's where the x Axis)

-0.25 x180 + Residual = -1.50x180
Residual = -1.25x180

Average residual astigmatism is -0.50x90 DC

What lenses would you give for the following situation:

WTR astigmatism with little or no sphere	RGP – Bitoric No: - Toric SCL
Low sphere huge WTR astigmatism and **needs an add.**	Monovision RGP With overspecs?
Normal script with astigmatism and needs an add	Proclear
ATR astigmatism	Lens are kinda limited, but soft CL is good because the thicker part of the len under lids RGP less then ideal because the thick part make sit harder to stabilize
Irregular astigmatism	RGPs
Keratoconus	RGPs
Low corneal cyl, high residual cyl situations	Front surface Toric → but go for SCL instead

Why would you not fit a high Dk Rigid material:

- Ex PMMA wearers, used to rough handling
- Ultra thin RGP, needs something of high r strength and lower dk, the t will compensate.
- Scleral designs or patients prone to hypoxia → go with XO2 with exceptional Dk/T
- Problems with wettability

Corollary question: when would you fit

- Core concept: Spectacle Rx = Tear film power + residual + BVP
- Core concept 2: back surface = fitting, however you can use tears. Front surface = power.
- Core concept 3: SAMFAP rule is done relative to the **flatter meridian.**

Spherical soft	Spherical rx, simple fit If the corneal is astigmatic but the prescription is spherical it is a strong indication for a soft spherical fit Compensate the rigid by using the **flatter** meridian
Toric Soft	>2.00 DC – first line, see if visual acuity is good. - average thickness is higher due to cylindrical and stabilization system Don't rely on centre.

Spherical RGP	Corneal cyl <2.00D, using tear film to mask. If most of this cyl is **corneal.**
	
	Note well: a spherical lens will **not** correct all the cylinder with the tears.
	
	How to order
	- BC: different principles: average, Flat + ¼ ΔK or 1/3 ΔK d
	  ○ Anymore then >3.00 some sources >2.00 → have to go back toric
	- BVP: Rx Sphere + TF factor (plus TF give more minus)
	  ○ Cylindrical component taken care of.

FST RGP	Spherical cornea but astigmatic prescription suggesting most of the correction is internal. With FST largely not useful since requires a good corneal fit and **stabilization system** You would use SOFT TORIC MOST OF THE TIME. FST BVP would be BVP from spherical fit + overfraction. How to order: - BC: as you would with spherical - BVP (sphero cyl): SAMFAP adjusted BVP + Over-refraction (in plus cyl)
BST RGP Example of utilization of BST induced cyl +1.75/-5.75x180 K: 42@180, 46@90 ∴ ΔK = -4.00x180 ∴ Res = -1.75x180 5.75/4.00 = 1.43 Induced = -2.00x180 Residual = -1.75x180 Induced will correct the residual	Good lock on the back – indicated in cases where theres high corneal cyl. **If you can understand the induced astigmatism,** you can use it to correct the **residual astigmatism** (if refractive = 1.2-1.4x corneal) *this is rare.* **If you can do it, it will be cheaper for the patient.** Corneal cyl x 1.2-1.4 = Refractive Cyl How to order: - BC: use the K measurements ± Mandell Moore Fitting factors (flatter) - BVP: SAMFAP Spherical (for any Mandell Moore adjustments) as the back surface corrects the cyl already (no TF, just lens) **Beware:** - BST needs a corneal astig to "lock" - Induces astigmatism due to tears ≠ lens in refractive indices **Induced Astigmatism** ≈ 0.5 x ΔK (approximate) Example. Patient has K values of 41 and 43 therefore their astigmatism is -2.00DC however induced will be -1.00DC extra. Your overrefraction will give you +1.00/-1.00x(other axis) **Induced astigmatism is inherent to the differences in BOZR**

<table>
<tr>
<td>Bitoric</td>
<td>
- >2.00 Corneal Cyl better optics then glasses or SCL torics
- BST didn't work because the corneal cyls x 1.2-1.4 ≠ refractive cyl

They have a toric front and toric back
<table>
<tr><th>SPE (spherical power effect)</th><th>CPE (cylindrical power effect)</th></tr>
<tr><td>Refractive = corneal cyl
(already corrected by BST)</td><td>Refractive ≠ corneal cyl
Need a residual correction</td></tr>
<tr><td>Used when you only wanna fix the induced cylinder of a BST</td><td>Fixes both induced and residual</td></tr>
<tr><td>Front cylinder = induced astigmatism</td><td>Front cylinder = induced and residual</td></tr>
<tr><td>Induced Cyl correction follows the axes so theres no need to worry about rotation</td><td>Residual may not be the same power so stabilization needed.</td></tr>
</table>

Specifying Bitorics:
Curves: give the BOZR on K for the alignment fit ± Mandell Moore
BVPs: give the power at each eye as according to ocular Rx, adjusting for flattening
Lab works out the rest. 2
Note if you want to adjust for the curves you have to SAMFAP each meridian.
</td>
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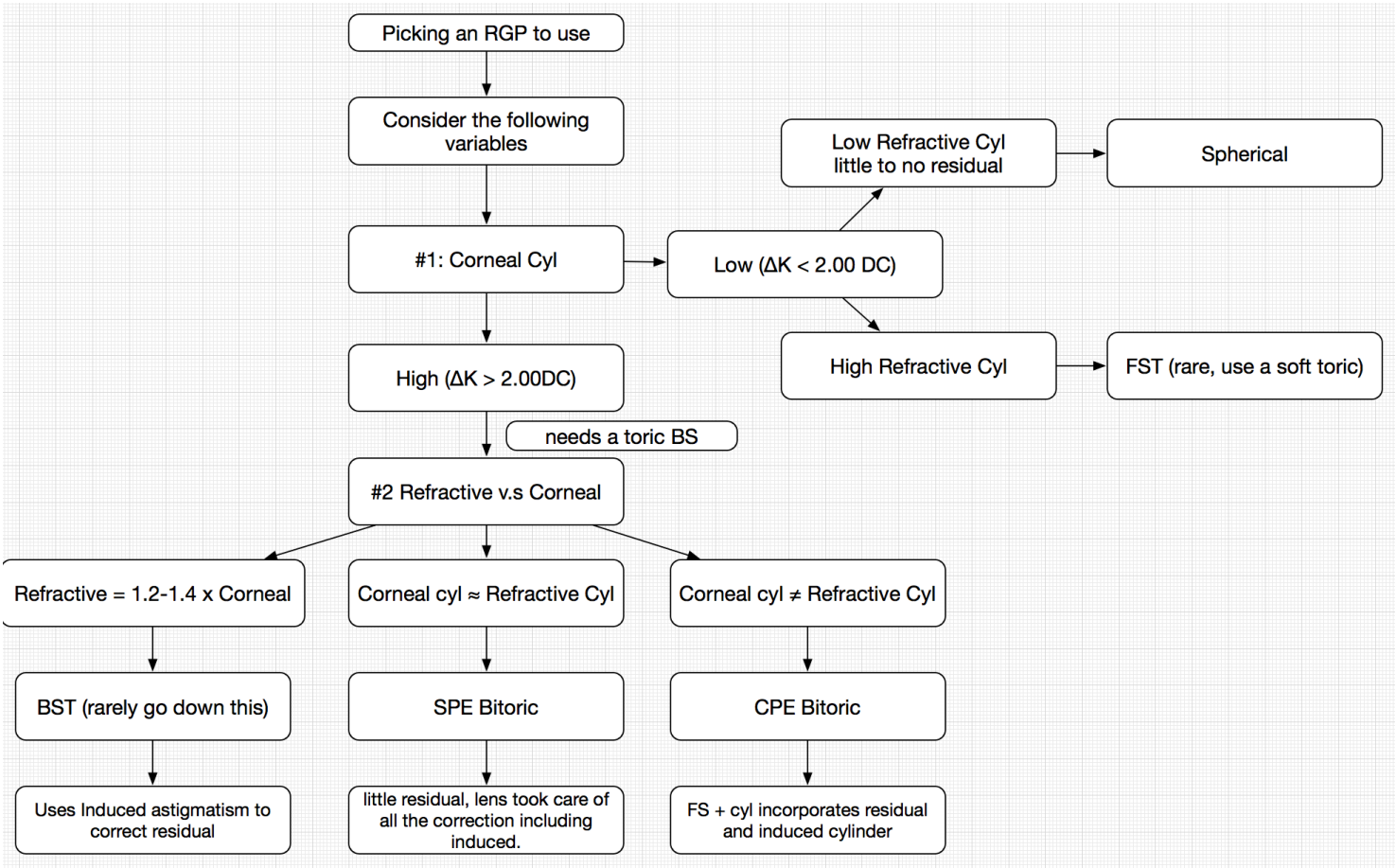


Problems soft and hard lenses.

Soft lens Discomfort	Hard lens Discomfort
SEAL Higher modulus lenses CLPC – protein CLPC – mechanical e.g. modulus Tight lens Loose lens Excessive edge lift, hyper mobile lens.	Signs of Spherical failure - Lens rocking around a meridian - 3 and 9 o'clock staining - Decentration - Lens flexure. What can you do about spherical failure - Go to a bitoric fit. - Aspheric fit - Smaller BOZD - Steeper BOZR Problems with optic zone and centration Too much movement Edges too thick/standoffish Poor wetting

Other problems

- **Insufficient tear exchange → flatten curves (BC and PC), blend junctions**
- **Seal off → widen peripheral curves.**
- **Dimple veiling → not true trauma, refit with steepper lens.**
- **Lens adherence: → refit and lubricants.**
  - **Nasal = flat lens**
  - **Inferior = tear evaporation**

Comparison charts of soft v.s. hard lenses

Soft lens benefits	Hard lens benefits
Initial comfort	Oxygen pump
Initial Cost	Long term cost
Cheap replacement	Less inflammatory side effects
Easier to fit	
Suitable for sports	
Doesn't interfere with CTT	

Lens troubleshooting think about the lens factors.

Positional problems	Staining	Halos/Flare	Early VA problems	Late VA problems
Problems: high, low, side by side, falling out, no movement. Special case: ATR lateral movement Modifiers - Curvature - Size - Zones - Weight - junctions - lenticulation B: parameters G: additions	Lens: Edge lift correction with peripheral zones Ill fitting lens Patient: MGD, Dry eye	Firstly, optimze the fit first Secondly consider the OZD + Pupil size Keratoconus.	Astigmatism → check over-refraction Wrong lens power → check for flexing	Warpage → verify Wrong lens worn Deposits → progent and polish

Case 1

History

- Aftercare
- **Fluctuating vision + Discomfort** → at the end of 8 hours wear.

Findings v.s. Interpretation

Findings	Interpretation
<u>Differentials for</u> Fluctuation of vision Lens discomfort	Sources of these symptoms include: - Steep fit - Flat fit - Uncorrected cyl + bigger pupils - Lens optics bad - Dry eyes
CL BC = 8.4 K = 7.6	K + 0.8 = BC, so calculation was done right. However after long wear the lens shrinks and can make the fit more steep.
Tight lens fitting Eccentricity? 0.5 (values 0.4 to 0.57) How does eccentricity work? E^2 = how many more mm flatter. e.g. e= 0.4 the periphery is 0.16mm flatter peripheral curve = 13-14mm radius	However the peripheral cornea and sclera are flatter then normal so lens is too tight. Consequences of tight lens: - SEAL - CLARE - Poor tear exchange - Indentation staining - Limbal vessel blanching, choking off blood supply.

Management for this case

- Check for any staining
- If there is lubricants
- Go to a flatter lens, 8.4 too steep, to account for lens steepening with dryness.

Case 2

History

- Extended wearer
- **Pain** → but no other signs of MK (e.g. blur, lid discharge, photophobia, redness)

Findings v.s. Interpretation

Findings	Interpretation
Pain → need DDX	- SEAL → pain + horizontal - CLPU → ulceration pain - CLARE → pain, but no redness - Toxicity → pain - SLK → pain - MK
Unremarkable fit	Probably not a mechanical cuase
PureVision 2 1.1 modulus myopia K = 8mm Lens 8.6 = steep lens Asian male	- High modulus 1.1 mPa - High edge thickness - Steep lens pushing and splitting the epithelium form the rub - tight top lid
Horizontal lesion	SEAL
EW	More likely to be SEAL since the cornea doesn't have time to adjust

Management:

- Take off contact lens
- Stop EW
- Lubricant every 2 hours – preservative free
- Consider prophylaxis if severe
- Refit into another lens… possibly a softer lens and flatter, no more EW

Possible questions:

- Would you give a steroid? No, not inflammatory.

Case 3

History:

- 1 month history of redness and irritation
- Been wearing lenses more then 10 years and wears then 12 hours a day.
- Current lenses are 1 year old.
- Solution is whatever she can get
- Oral contraceptive

Findings	Interpretation
Recent history	Something has changed, be it systemic or something to do with compliance
Long time wearer	Not necessarily compliant careful
Current lens 1 year old	Consider: - Protein deposits - Lens damage
Unsure about cleaning solution	So they're NOT compliant: - Could be preservatives - Could be bad neutralization
Oral contraceptives	Dry eyes and systemic variations
Wearing specs to consult	Contact lens intolerance
High rx	Thick lens → seal?
Fitting = slightly flatter	That's ok, it's a hydrogel
Staining	Superior… SLK, SEAL – depends on the picture
Redness	Only superior!!! Strongly consider SLK
Papillae	Ddx: CLPC – but SLK indicated too. **However, if the papillae are on the cornea → definitely consider SLK**
Thick protein film on lens	Could be protein reaction CLPC

Differentials

- SLK
- CLPC – really close however CLPC: **does not have corneal staining**
    - Protein deposits
    - Ripped lens, mechanical cause
- Lens damage
- Solution toxicity
- Non neutrolizing
- Solution problems
- Seal
- Dry eyes from oral contraceptives

Final dx:

- SLK, given that theres superior staining and papillae

Management

- Discontinue lenses, lubricants 1 week review.
- FML 0.1% BID (questionable – but in the slide nonetheless)
- Refit into fresh lenses - monthly lenses since full time wearer
    - I would go Biofinity as it's a silicon hydrogel, more oxygen

- Change to preservative free solution
    - I would go AOSept
- SLK also has inplications with **thyroid** conditions.

Discussion points relevant to this case

Key diagnostic signs:

- Superior limbic appearance - papillary response
- Localized of bulbar injection and corneal staining
    - If preservative → should be more diffuse and bilateral symmetrical.

Refitting considerations - Ionic v.s. non-ionic

- Going to a Silicon Hydrogel is less water.
- Water is a more ionic material
- Protein is ionic – more attraction in hydrogel lens.
- It's more sensitive the lens care system too (retention in the matrix)

Why did we have ionic material

- Increasing wettability of lenses, originally thought to increase oxygen through water.
- However later found silicon based better at conducting oxygen
- Balance between lens wetting and oxygen.

SLK:

- Source: dirty lenses or CL SLK
- Common offending agent is Thiomersal, can be found in saline→ best to go AOSept

Go learn how to differentiate SLK and Theodore's SLK

Theodores has an association with thyroid:

- Usually not wearing CLs
- Less severe
- Less palpebral conj
- **Filamentary keratitis**

Case 1: AB blurry and uncomfortable SCL

Hx

AB, 30 year old Asian male presents for 1 week Contact lens aftercare. Complains of fluctuating vision and lens discomfort towards the end of the day with about 8 hours of wear time. Optifree MPS used. POHx and PGHx unremarkable.

Clinical findings

- CL parameters
  - Oasys 8.4/14/-3.50 OU
- CLORx and VA
  - OD: pl/-0.25x90 (6/7.5)
  - OS: +0.25/-0.25x90 (6/7.5+)
- Retinoscopy reflex slight distortion centrally
- CL fitting
  - Central position
  - Minimal on primary gaze blink
  - Minimal lateral lag
  - Minimal movement on push up
- K readings
  - OD: 7.71/7.58@90
  - OS: 7.63/7.58@90
- Subjective refraction
  - OD: -3.50/-0.25x90 (6/6)
  - OS: -3.25/-0.25x90 (6/6)

Questions

- DDx?
  - Please Justify each DDx
- Final Dx?
  - Please justify your final Dx
- Mx Plan?

s

Case 1: AB blurry and uncomfortable SCL (discussion points)

DDx

- Uncorrected astigmatism
  - Can you get SCL torics in -0.25 steps?
  - Does it fluctuate?
- Defective lens optics
- **Steep fit**
- Flat fit
  - Does it make sense with the actual CL fitting observation?
- Dry eye

Actual Mx

- Refit with flatter BC
  - Oasys 9.0/14.0/-3.50 OU
    - BCVA improved to 6/6 each eye
    - Also increased movement in fitting leading to increased comfort throughout the day
    - Able to wear more than 12 hours a day

Discussion

- The initial BC was far flatter than actual K readings, why do we need to refit with even flatter lens?
  - Peripheral cornea and sclera flatter than normal then the lens will fit too tight
    - *What is the average e values of peripheral cornea?*
      - *Between 0.4 and 0.57*
        - *Lower more spherical*
      - *Topographer does not disclose the distance from centre and it takes average so just gives an indication of rate of flattening*
        - *Eg 0.4 e value means periphery 0.16mm (square of 0.4) flatter*
    - *Average radii of curvature of sclera close to limbus*
      - *Between 13.3 to 14.3mm*
  - Tight fitting lens leads to poor comfort and vision
    - Poor comfort;
      - Decreased lens movement leads to poor tear exchange, lens binding and hypoxia
        - Long term effects: CLARE and infiltrates
    - Tight lens FL stains;

Case 2: CD Pain with SCL wear

Hx

CD, 40 year old Asian Male with 1 day Hx of moderate sharp pain OD. No blurry vision, lid discharge, photophobia or redness. Been fitted 2 weeks prior with Purevision 2, has been wearing as extended wear for 7 days. On 1 week CLAC the findings were unremarkable, the lenses were removed in the room and disinfected and reinserted. Since then he has been wearing it for 3 days until Sx arose. Previously he was wearing daily disposables with no problems. POHx and PGHx is unremarkable.

Clinical findings

- CL parameters
  - Purevision 2 8.6/14/-3.75
- CLORx and VA
  - OD: PL (6/5)
  - OS: PL (6/5)
- CL fit:
  - Well centered, less than 1mm on blink primary gaze, good push up movement, 1mm lateral gaze movement OU
- Anterior health:
  - OD: superior bulbar conjunctiva injection, horizontal corneal epithelial lesion parallel to limbus which stains with fluorescein (see photo)
  - OS: unremarkable
- K readings (prior to initial refit)
  - OD: 8.13/7.99 @ 90
  - OS: 8.08/8.04 @ 90
- Subjective refraction (prior to initial fit)
  - OD: -4.00/-0.50x180 (6/5)
  - OS: -3.75/-0.50x180 (6/5)

Questions

- DDx?
  - Please Justify each DDx
- Final Dx?
  - Please justify your final Dx
- Mx Plan?

Case 2: CD Pain with SCL wear (Discussion points)

DDx

- Superior limbic keratitis (SLK)
- **Superior epithelial arcuate lesion (SEAL)**
- CLARE
- SCL hypoxia
- GPC
- Solution preservative toxicity

Actual Mx

- Cease CL use until lesion healed
  - Preservative free lubricant q1h
  - FML bid
    - Steroid necessary?
- 1 week review the lesion was healed completely
- Refit into flatter SCL: Oasys 8.8 BC

Discussion

- SEAL is essentially epithelial splitting from mechanical rubbing from EW of tight lens
  - Ie Chronic indentation superior peripheral cornea by peripheral curve junction of SCL (explaining horizontal orientation to limbus)
    - Exacerbated by high peripheral thickness and high modulus
  - Long term wear leads to adjacent conj injection
  - Non-inflammatory in nature but epithelial hypertrophy may result giving that heaped up appearance
    - In rare cases pannus appearance results
- Mx
  - Resolves 1-2weeks with cessation of wear
  - Changing to thinner periphery or lower modulus or flatter curve alleviate peripheral compression
- Was the patient appropriate candidate for purevision 2 given flatter Ks and tight lids in nature?

Case 3: EF SCL discomfort

Hx

30 year old Caucasian female 1 month Hx of redness and irritation in OD. Using rewetting drops and discontinuing SCL wear reduces Sx but resumption of wear causes Sx to return. Px wears conventional SCL 12 hours a day and worn lenses for more than 10 years, current SCL is almost 1 year old. Px unsure of cleaning solution used and just gets what she can from pharmacy. Other than taking oral contraceptives, POHx and PGHx unremarkable. Px presented to exam wearing spex.

Clinical findings

- Subj Rx and BCVA
  - OD: -5.50/-0.50x10 (6/6)
  - OS: -4.25/-0.50x30 (6/6)
- Topography
  - OD: 7.42/7.34 @ 80
  - OS: 7.25/7.34 @ 80
- Anterior health
  - OD: G3/4 superior bulbar conj injection, G2/4 papillae, G2/4 superior corneal staining, clean a/c (refer to photo)
  - OS: G2/4 papillae, clean a/c
- Lens inspection
  - Thick protein film OU with dry patches; no mucin balls
- Habitual CLRx; 55G (hydrogel Monocurve design 55% water content)
  - OD: 8.6/14.0/-5.25
  - OS: 8.6/14.0/-4.25

Questions

- DDx?
  - Please Justify each DDx
- Final Dx?
  - Please justify your final Dx
- Mx Plan?

Case 3: EF SCL discomfort (Discussion points)

DDx

- GPC
- Preservative sensitivity
- **SLK**
- Episcleritis
- SEAL

Actual MX

- Cease CL wear until redness resolved
  - Preservative free lubricant q1h
  - FML bid
- 1 week redness resolved
- Refit to disposable SCL
  - B&L Ultra 8.5/14.2 OD:-5.25 OS: -4.25
- AO sept solution as cleaning regimen

Discussion

- Giveaways of Dx
  - Localisation of bulbar injection and corneal staining to superior quadrant
  - Protein debris on lenses
  - SL papillary reponse
- More or less immunological response to CL with protein coating and preservatives
- Solution preservative sensitivity ruled out as it would manifest as
  - Bilateral
  - Diffuse redness and itchiness
- Differentiated from Theodore's SLK:
  - CL induced has more intense SL reaction
  - Theodore associated with:
    - Hyperthyroidism
    - 20-50 year old women
    - Comes and goes over 5-10 year period
- Once redness resolved:
  - Refit with disposable lens with low water content and non-ionic surface recommended
    - Difference between ionic and non-ionic?

- Ionic material added to increase hydration, thus makes them more sensitive to lens care system and increase uptake of lysozyme and other positive charge particle in ocular environment
- What is thimerosal?
  - Can be found as preservative in saline solutions
  - In old lenses with protein debris can irritate the eyes

Clinical Pearls

- CL-SLK should be differentiated from Theodore's SLK
- CL-SLK can be caused by solution preservatives or dirty lenses
- Thimerosal can be a common cause of CL-SLK

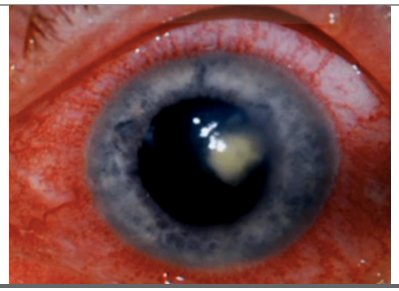
Area 5: Adverse effects and CIEs → from year 4 *(theres no easy way to do this area)*

General Contact :

what lenses are they wearing	How long each day	How long today	What solutions used, show me the compliance, show me how to do it
How long have they worn CLs for	Any lapses in compliance	Past events/had to stop wearing lenses	how does the lens feel at the end of the day.
If I could do no more would you still take these lenses	Do you love your contact lenses	Any problems in the distance/reading?	What type of sensation

Template for Condititions

Disease Entity (name)	**Picture**	**Contact lens questioning**	
Signs		Symptoms	
Management		Other Info	

<table>
<tr><td>Microbial Keratitis
Differentials:
CLPU
Fungal
Acanthomeba
CLARE</td><td>  </td><td>Contact lens questioning</td><td>➔ Have you removed the lens? Did it make a difference? No</td></tr>
<tr><td>Signs</td><td>- Central large lesion → (won’t change as much in early disease)
- Diffusely red
- Epithelial defect
- A/C involvement
- Lid Involvement
- Lowered VA
- Hypopyon</td><td>Symptoms</td><td>- PAIN
- Blurry vision
- Photophobia
- Lots of pain
- Discharge: variable mucous, pus, tears.</td></tr>
<tr><td>Management</td><td>1. Loading dose q15m/6hrs
2. Q30m rest of day
3. Q1hr next day
4. Q4h next 14 days
Next day review → 3 days → weekly.</td><td>Other Info</td><td rowspan="2">Tissue is necrotizing
- Most causative organism shown to be P.A.
Steroid use controversial, use only if epithelized. NOT IN EARLY DISEASE

Scrape → Blood Agar and PCR

Differentials:
Fungal → feathery margins, AC reaction,
Acanthamoeba → pain, disproportionate, severe.</td></tr>
<tr><td>Hospital management if severe or non responsive to your treatment</td><td>1. Dual fortified Tobramycin 1.5% q10m/2hrs then every hour after that
2. Cephazolin 5% every hour in between Tobra
3. Night chloramphenicol or tobramycin ointment</td><td></td></tr>
<tr><td>Differentiation of MK v.s. CLPU</td><td colspan="3">Infiltrate size (CLPU smaller), location (CLPU more peripheral) and shape (CLPU rounder), density, pain, intensity of redness. Lid and AC involvement. Corneal odema, discharge. NO IMRPOVEMENT ON LENS REMOVAL.</td></tr>
</table>

Indications for referral for fortified:

- Deep
- Big
- old
- Monocular
- In the visual axis

Why is PA so aggressive:

- protease
- biofilm
- increased resistance

severe disease:

- environmental organism
- tropical: humid and hot
- delay in seeking treatment
- smoking, SES, case, overnight and solutions

implications with susceptibility

- **IL6** increases elimination of pathogens (less susceptible) and decreases inflammatory effects (less severe)

What if its Acanthamoeba:

- Disproportionate, low signs, high smyptoms. Takes long, NO STEROIDS. Water related.

CLPU (recurring)	**Picture**	**Contact lens questioning**	**Extended wear?** **Any pain?**
Signs	Deep scar (but more peripheral) Smaller scar Redness Watery → Purulent =MK AC Clear (unlike MK) Vision clear (unlike MK)	Symptoms	Pain (ddx MK) Discomfort
Management	1. Stop lens wear 2. Monitor closely for 24h → in case MK 3. Lubricants 4. If suspicious have Cipro QID cover 5. If severe inflammation have Cipro and Flarex QID	Other Info	Causative organism: Staph aureus (+) **but it is an inflammatory reaction.** Is a sterile condition which may have involvement of some kind of trauma

CLARE (recurring)	**Picture**	**Contact lens questioning**	**Extended wear? Slept in lenses** **Any pain?**
Signs	Bright red eye Infiltrates	Symptoms	Pain (ddx MK) Remove lens → relief

Management	1. Monitor 24 hours 2. Lubricants 3. If patient concerned → Flarex 0.1% QID 4. Look for epithelial defects. 5. No EW. 6. Fit into fresh new lenses	Other Info	Associated with H. Inlluenzae (gram negative) – endotoxin. Also associated with streptococcus pneumoniae and staphylococcus aureus. Closed eyes → releases endotoxins → inflammatory cascades Redness goes away in days.

IK (infiltrative keratitis)	**Picture**	**Contact lens questioning**	**Extended wear? Slept in lenses Any pain?**
Signs	Red eyes (not as severe as CLARE or MK) Infiltrates → more towards limbus Can have staining but not thick. Some discharge	Symptoms	Irritation (but no pain)/ FBS.
Management	1. Stop CL wear 2. Lubricants/washes/cold compress 3. If patient concerned Flarex 0.1% QID 4. Review 24-48 5. AC reaction → cycloplege but RARE. IF YOU CAN'T RULE OUT RISK OF INFECTION, PROPHYLAXIS	Other Info	IK just like CLPU is associated with staph, **could be a relationship with Bleph/marginal – so manage those conditions!** **Infiltrates are a collection of PMNs in the cornea as a response to some form of bacteria but no infection.**
Risk factors for CIE	EW, epithelial break, contamination, hygiene, internet, smoking, new weaer, non compliance, environment. Silicon hydrogel, younger, Male, higher Rx, Sick conditions		

General inflammation reduction measures

Contact lens	Technique and hygiene	Health of patient
Change to EW Change to DD or a lower frequency Change to RGP Lower modulus	Keep the case clean Rub and rinse Avoid preservatives → go with Peroxide system Add a protein step	Manage any MGD or blepharitis

SICS	**Picture**	**Contact lens questioning**	**What did you do this time? What solution?**
Signs	- Diffuse staining -	Symptoms	Pain, irritation, patient usually knows
Management	- Stop lens wear - Lubricants - Ensure compliance - Change to a non preserved system - Review lens and solution combinations	Other Info	Peroxide in the eye? - Irrigate profusely - Short course of FML 1-2 days QID - Short course of NSAID as alternative if HSV

Corneal Erosion	**Picture**	**Contact lens questioning**	**Injuries to the eye?**
Signs	- Bright staining - Negative staining at times due to the "mound" - It's the opposite of CLPU, theres no infiltrate but staining	Symptoms	Moderate to severe pain Watery eyes
Management **Cycloplegia generally:** **Cyclopentolate 1% BID**	1. Stop wearing CL s 2. Prophyl Cipro until re-epithelization + 3 days after. 3. Pain management since no **steroids → ketorolac 0.5% QID or oral** analgesis 4. Review in 24 hours	Other Info	**No steroids (no infiltrate)** NO PATCHING

SEAL -	**DDX SLK look for:** - Allergies - Solution - Papillae - Thiomerosal	**Contact lens questioning**	**Length of wear** **What lens is worn** **How does the lens feel**
Signs	- Arcuate lesion 1-2mm - 10-2 - Little inflammation	Symptoms	Asymptomatic, mild pain
Management	1. Discontinue CL wear 2. Lubicants 3. Prophylaxis if severe. 4. Refit into softer lens or RGP	Other Info	- More silicon hydrogel - More tight lens - More so if high modulus lens - More so if toric lens

VLK	**DDX:** **Phlyctenulosis** **Pinguecula** **Ptergium**	**Contact lens questioning**	**Not much really**
Signs	Symptoms	Management	See below
SPK	Asymptomatic	Lubricate and fit	Limbal red spot finlamamtion, associated with 3-9 staining caused by lens edge profile and edge lift insufficiency
Coalescing	Increased lens awareness	Stop lens wear 5 days	
More staining vascular tuft	Reduced wear time	FML and discontinue	
Increased signs	Now very painful	stop lens wear for 3 weeks Steroid (FML) + AB (Cipro)	

CLPC		**Contact lens questioning**	**Blinking feeling?**
Signs	Papillae under lids	Symptoms	Reduced lens wear time Itching Mucus
Management Therapeutic management → **for moderate to severe** **1. Suspend wear (weeks to months)** **2.** Add agents: - Olopatadine 0.1% or - Ketotifen 0.025% BID **3.** can use steroids short term FML 0.1% QID about 1 week → **taper quickly**	1. Document grade the size, redness, zones, Local → mechanical Diffuse → immunological Papillae stains 1. Stop lens wear 2. Saline and cold comp 3. Review compliance 4. Add protein remover 5. Refit lenses a. Lower modulus b. Daily disposable c. Go to SiHy d. Less wear time Note can reintroduce lens wear even if not fully resolved.	Other Info Sources - Mechanical rubbing - Deposits - Hypersensitivity - genetics	Risk factors: - EW - Wear time - Deposits - Allergies - Larger lenses - MGD CLPC: olopatadine was found to be just as effective as FML

Injury management:

How long does it take for epithelium to heal

- Complete turnover = 7-10 days
- *Superficial cells are non-keratinised squamous* cells which are attached to the glycoalyx, a prevention of binding of pathogens. They're held together by desmosomes and tight junctions made up of zonula occludens. This is a mechanical barrier.
- 2nd layer = wing cells. Desmosones and aherens junctions. Has gap junctions and demosmosomes to allow for molecular exchange.
- Basal layer = columnar cells. has gap junctions and desmosoems as well.
- **Basement membrane** plays a role in healing, growth factors etc.

Phases in wound healing

1. Latent, cellular remodelling and changes in *tear composition for cell healing.* This increases productions e.g. MMP-9s which degrade the basement membrane so that the cells can be remodelled **this step happens over several hours** and involves cells to apoptose and shed into the tear film. Gap junctions open up to allow for cells to past through, basal cells develop feet and…. → if it is only minor 3-6 hours.
2. Migration **24- 48 hours** – cells go to the wound edge and spread to the denuded area until its fully sealed

At this time the tight junctions are not fully sealed..

3. Proliferation – the monolayer starts to reproduce and fill up. Note at this stage, the cells aren't attached and are still vulnerable. **it takes up to a week for reattachment.**
4. Reattachment: anchoring cells attach to the substrate with hemidesmosomes. If the basement membrane didn't have any damage then it's full recovery is measured in days however if it did, then re-anchoring takes
    a. This is why you check over a weeks time
    b. This is why RCEs are hard to treat.

What's the impact of diabetes on Cl wear?

- Dry eye - a sign of neuropathy
- Altered tear chemistry
- reduction in corneal sensitivity
- poor wound healing due to impaired inflamamtry cell function.
- Endothelial differences

→ **needs more oxygen.**

What would you do if patient was on other drops e.g. anti-allergy and why would you do it
this way.

- Anti allergy = CLPC risk → continue the anti-allergy drops, monitor more carefully..
    - Zaditen minims are good.
- Put them in before the contact lens so that it all gets in.
- Drops and preservatives → watch the preservatives going into their eye.
- If they have preserved lubricants, move to Preservative Free
- Go with a silicon hydrogel daily disposable

Possible effects of systemic medications and general health status on contact lens wear

CONDITION	POTENTIAL PROBLEM	MEDICATION	POTENTIAL PROBLEM	ADVICE
Allergies	Preservative reactions Atopic reaction to deposit build-up	Antihistamines	Atopic conjunctivitis Contact lens associated papillary conjunctivitis Reduced contact lens tolerance	One-day disposable contact lense or non-preserved systems with frequent replacements lenses Monitor – check for dry eyes
Skin condition (eg eczema)	Excessive deposits Lid irritations Blepharitis Punctate keratitis			One-day disposable or frequent replacement lens Thin edge design Avoid contact lens wear until clear Monitor – do not fit if clinically significant
Diabetes	Reduced epithelial healing			Close monitoring
Thyroid dysfunction	Tear deficiency and poor blinking			Avoid contact lens wear
Vitamin A deficiency	Mucus deficiency deposit build up			Monitor – possible soft frequent replacement
Systemic hypertension		ß-blockers, diuretics	Dry eye	Monitor
Psychosis	Contact lens adaptation	Psychotics	Dry eye	Monitor – contact lens wear possibly contraindicated
Hormone changes eg birth control, pregnancy, menopause	Dry eye Corneal contour changes Changes in corneal sensitivity	Oral contraception	Dry eye	Monitor

SCUT study and steroid use

Srinivasan 2015

- Adjunctiva topical corticosteroids associated with long term bacterial cornal ulcers that are **not caused by nocardia species**.
    - For this reason you should have a culture proven.
- Might be useful with pseudomonas infection → important for contact lens.
    - Overall no improvement
    - But the invasive subtype showed 2.5 line improvement v.s. Placebo
- Conclusion:
    - No improvement in spectacle acuity at 3 months
    - Despite this, subgroup analyses found that it was helpful in:
        - Low vision groups count fingers initially.
        - Big ulcers covering pupil 4mm
        - Patients with deeper ulcers.
    - But didn't lead to more adverse effects

- Cochrane review concluded not enough evidence to use steroids however Cochrane only reviewed 4 studies but only SCUT was sufficiently powered

Important study details:

- 50% streptococcus pneumoniae
- 25% pseudomonas aeruginosa
- 10% nocardia species → however this is very rare in US, its more india (where most of the study was conducted)
- The study indicated that steroid treatment in 2-3 days had better result then those with delayed treatment
- Study parameters aimed for 48 hour course.

Other notes:

- When you use steroids – it increases IOP and also drags the inflammation duration.
- AAO recommends steroids 48 hours after, if you know the culture.

Area 6: Professionalism and Clinic flow

Patient presents with a CL problem

- History
  - CC – history as normal
  - CL
    - Type of lens
    - Schedule
    - Replacement frequency
    - Cleaning regimen
    - Symptoms over time
    - Past events
- Exam:
  - With CL in (white and fluro)
  - Without CL (white and fluro)
  - Evert lids
  - Grade the disease.
- Thinking:
  - Infectious or inflammation → consider scrape
  - Drugs: not chloramphenicol, never below MIC
  - Any comorbidities
  - NO BANDAGE CLs in CL WEARERS (CLs themselves are low grade trauma)
  - NO PATCHING
  - Not always O2 related
- Management points
  - Discontinuing lens wear
  - Changing lens factors → disposal frequency
  - Changing hygiene factors → solutions, types,
  - Changing patient behaviours

Preliminary contact lens examination

PATIENT DISCUSSION	PATIENT EXAMINATION
Personal details • Name, address • Age • Dexterity • Complexion • Occupation, recreation **Contact Lens specific** • Expectations • Motivation **General History** • Systemic disease history • Familial disorders • Motivation **Ocular history** • Refraction correction past • Ocular disease history • Ocular symptoms	**Full eye examination** • Vision • Refraction • Binocular assessment • Ophthalmoscopy **Measurements** • Corneal curvature • Corneal diameter • Pupil size • Palpebral aperture **Examination of anterior segment** • Lids and lashes • Conjunctiva • Tears • Cornea epithelium stroma endothelium
ASSESSMENT OF SUITABILITY	

You have a patient in your chair who is my prescription (-3.75/-0.75 x 45) now what.

- History
- Refraction confirmation → if pertinent
- K readings and Topo if possible
- Health Check
- Put the lens on
- Over-refract
- Check lens fitting
- Give the lens to trial
- Aftercare in a week.

Area 7: Drugs → go to thera notes as well as the

https://www.catalogue-lentilles.fr/wp-content/uploads/2015/01/Staining_Grid.pdf

Chlorampencol			
Indications for use	Conjunctivitis, blephariti, foreign body, prophylaxis, SPK Non contact lens case.	Dosages and percentages	Comes in ointment 1% (use as adjunct) 0.5% drops and minims
		MIC	4x/day
Mechanism of action	Binds to 50S subunit to inhibit protein synthesis	Special populations	Safe pregnancy Safe breastfeeding
Warnings and contraindication	**Not effective against pseudomonas aeruginosa.** (-) Blood dycrasias (-) bone marrow (!) FMhx of above	Other information	**Thermoabile** 1 month use. AC absorption. Grey baby IV/Oral only. Due to liver enzymes not developed.

Tobramycin			
Indications for use	Bacterial Keratitis Conjunctivitis	Dosages and percentages	Tobrex/Genoptic 0.3% 5ml Fort = 1.3/1.5 respectively
		MIC	QID (framycetin 3x)
Mechanism of action	30S subunit of ribosomal RNA. Protein synthesis inhibition. Bacteriocidial	Special populations	Safe for pregannce and kids
Warnings and contraindication	(-) Kidney problems (-) hearing loss (-)neuromuscular dz (!) allergic (!) inactivated by purulent discarges. Rare: delayed wound healing and retinal toxicity. Common: SPK, irritation	Other information	Targeted for gram negative but work for gram positive too Has a post antibiotic effect with P.A. Works in tandem with B-Lactam drugs but stagger so they don't get inactivated. Poor corneal penetration

Ciprofloxacin			
Indicatiosn for use Bacterial keratitis and Conjucntivtis			
Regimen	<u>Cipro</u> 6h: q15m Rest of day: q30m Day 2: q1h After: q4h re-epithelization Conjunctivitis: Day 1 and 2: Q2h After: Q4H Deep corneal ulcer: 500mg oral but would probably refer.	Dosages and percentages	Cipro = 0.3% Ofloxacin = 0.3%
Mechanism of action	Nuclear acid synthesis inhibitors, blocking DNA gyrase	Special populations	Ok for pregnancy and breastfeeding.
Warnings and contraindication	(!) watch out for corneal perofration… effect of MMP of collagen. (!) tendons ruptures – MMP (!) oral quinolones and NSAIDs Causes ocular irritation. Infrequently can make keratitis worse.	Other	
Other information	Ofloxacin - Better corneal penetration - More risk of perforation Ciprofloxacin - More precipitates		

The alternative:

- Cephazolin 5%
- Gentamicin 1.5% or tobramycin 1.3%

1. loading dose for q10m 1-2 hours of gentamicin
2. alternate every half hour:
    a. cephalosporin (on the half hour)
    b. gentamicin (on the hour)

Atropine or cyclopentolate 1% twice a day for pain amangement

Mono v.s. fortified

Mono	Fortified
- less hospital stay/duration of hospital visit - more serious side effect.	- go if deep, large, old - these are more likely to develop the

Steroids

Steroids indication to use:

- uveitis
- epi/scleritis
- allergic conjunctivitis/CLPC
- MGD/DED if severe.
- Cataracts
- Optic neuritis.

Adverse effects:

- Cushin syndrome type stuff
- Ocular hypertension, reversible.
- PSC if long term high dose.

Mechanism

- Kills the receptor mediated inflammation cascade, fibroblast and keratocyte activity.
- Acetate = lipophilic
- Phosphate = hydrophobic

Careful if:

- HSV history
- Glaucoma, history of steroid response, OHT, Narrow angles → mechanism uncertain but due to trabecular build up or cross linking. **1 in 3 people** responde to steroids. Glaucoma patients have a higher risk of being steroid responders.

Steroids ranking in terms of potency.
PredForte → Dex → Flarex → Pred Minims → FML/FLucon → Hycor
Potency is proportional to IOP rice potential

prednisolone			
Indications for use	Uveitis, episcleritis, resistant ocular allergies, post op	Dosages and percentages	Prednefrin 1% with 0.12% 0.5% minims (pre alone)
		Maintainance dose	QID
Mechanism of action	Inhibit phospholipase before AA gets converted to other stuff.	Special populations	
Warnings and contraindication		Other information	More intense treatment = every hour.

FML and Flarex			
Indications for use	FML – more conj stuff Flarex – corneal.	Dosages and percentages	Flarex = 0.1% FML = 0.1%
Mechanism of action		Special populations	
Warnings and contraindication		Other information	

Other steroids:
Hycor (hydrocortisone 1%) (preservative free) = lid/contact dermatitis
Maxidex (dexamethasone 0.1%) = allergic conjunctivitis and post op – poor penetration

1-4 hours: q15
Depending on severity after
Q1h= severe grade 4
Q2h = moderate grade 2-3
Q4h = mild grade 1

Then taper, longer taper if:

- Longer period of treatment
- Stronger drug
- More frequent delivery

Cycloplegics

Atropine and Cyclopentolate – indicated in uveitis and in MK			
Dosages and percentages	Atropine 1% 1-2x/day Cyclopentolate 0.5% 1% 2x a day		
Atropine 2x a day	Cyclopentolate 2x		
Mechanism of action	Blocks action of aceytlcoline the NT because they relax ciliary spasm by paralysing muscle, they prevent PS and stabilize the BAB to reduce cells/flare.	Special populations	Pregnant ok
Warnings and contraindication	Glare Accommodation blur ! head injuries ! angle problems ! lens problems ! IOP problems	Other information	Cycloplegic effect Cyclo = 6h to 1 day Peak at 25-75 mins Atropine 7-14 days, peak at 1-3h

Note: cycloplegia done a lot faster because drugs get broken down faster.

Pain:

- Result of signalling of inflammatory pathways
- Manage with analgesics
- Combine stimulants, anti-convulsants,

Olopatadine - patanol (S4) 0.1%
Safe to use for pregnancy and children
Hyperemia, keratits and dry eye – due to BAK probably.

Mechanism	Anti-allergic. Mast Cell stabilizer and H1 Selective antihistamine
Indications	CLPC, allergic conjunctivitis.
Other notes	Just as useful as FML in controlling CLPC but if its severe consider adding combo therapy Patanol more efficacious But ketotifen is S3 OTC.

Zaditen (Ketotifen 0.025%)

Mechanism	Anti-allergic. Mast Cell stabilizer and H1 Selective antihistamine
Indications	CLPC, allergic conjunctivitis.
Other notes	Comes in preservative free unit dose too. Watch out for overdosing, can worsen allergy.

Common optometric Products

Product name	Sterilid	Indications for use	Blepharitis
How to use the product			
Product specific info/Key Components			

Product name	Ocusoft lid wipes	Indications for use	blepharitis
How to use the product			
Product specific info/Key Components			

Product name	Ocusoft lid scub plus	Indications for use	Blepharitis
How to use the product			
Product specific info/Key Components			

Product name	Systane balance	Indications for use	Dry eye – aq def general purpose
How to use the product			
Product specific info/Key Components			

Product name	Systane Ultra	Indications for use	Dry eye – DED/MGD
How to use the product			
Product specific info/Key Components			

Product name	HyloForte/Fresh	Indications for use	More severe dry eye, more symptomatic
How to use the product			
Product specific info/Key Components			

Product name	Optifree Pro	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Total Care 1 Abbott	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	MD Eyes AREDS 2	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Reclens Saline	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Menicon Spray and Clean	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Menicon Progent	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Menicon MPS	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Progent 7 treatments	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Boston Advanced Cleaner	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Boston Advanced Comfort formula	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Boston simplus	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	Boston One Step	Indications for use	
How to use the product			
Product specific info/Key Components			

Product name	AOSept	Indications for use	
How to use the product			
Product specific info/Key Components			

BAK – destabilizes tear film
CHX – never used alone
Thiomerosal – fungi
Polyquad – bacterial, maybe fungal
Aldox – fungal

Polyquad – Optifree
PQ + ald**o** = everm**oi**st
PQ + **ale**xidine = revit**ale**ns
PQ + PHMB (**b**iguanide)= **b**iotrue

22.3.2 Advantages of soft and hard lenses in extended wear

Advantages of soft lenses

- More comfortable.
- Easier adaptation.
- Easier to fit.
- High *Dk*s with silicone hydrogels.
- No problems with foreign bodies.
- No 3 and 9 o'clock staining.

Advantages of hard lenses

- Very high *Dk*s possible.
- Tear pump on blinking.
- Avoidance of trapped debris.
- Reduced risk of infection.
- Better vision.
- Fewer corneal changes.
- Lenses do not cover the entire cornea.
- No lens dehydration.
- Fewer deposit problems.
- Easier lens maintenance.
- No solutions sensitivity.
- Longer life span.